Amazon Supply Chain Intelligence

AI-Driven Delivery Delay Risk Prediction Report

1. Executive Summary

This report details the performance and insights of the AI-driven delivery risk prediction system. The system utilizes advanced machine learning algorithms to predict whether a shipment will be On-Time, At Risk, or Delayed based on operational factors like traffic, weather, and distance.

2. Exploratory Data Analysis (EDA)

Before training, we analyzed the dataset to understand key drivers of delay. The following visualizations highlight the relationships between variables.

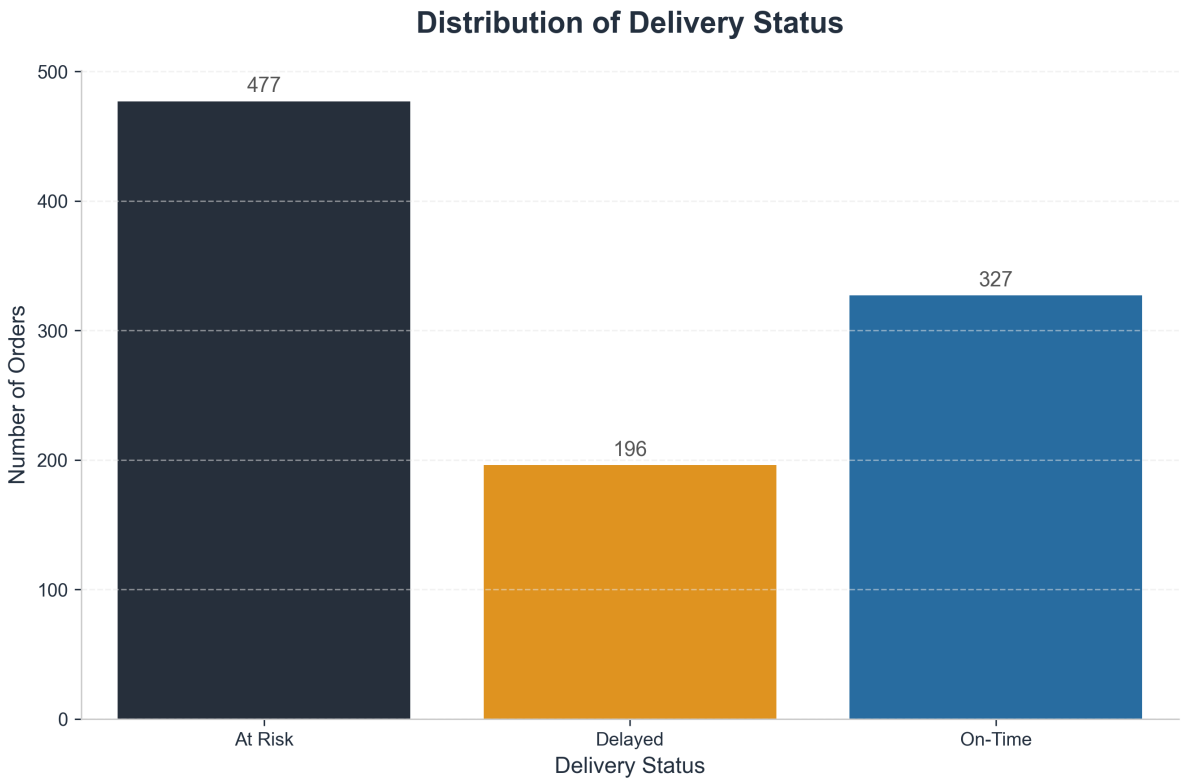

Fig 1: Distribution of Delivery Status Classes

This chart shows the balance of our training data. We used SMOTE (Synthetic Minority Over-sampling Technique) to ensure the model doesn't become biased towards the majority class.

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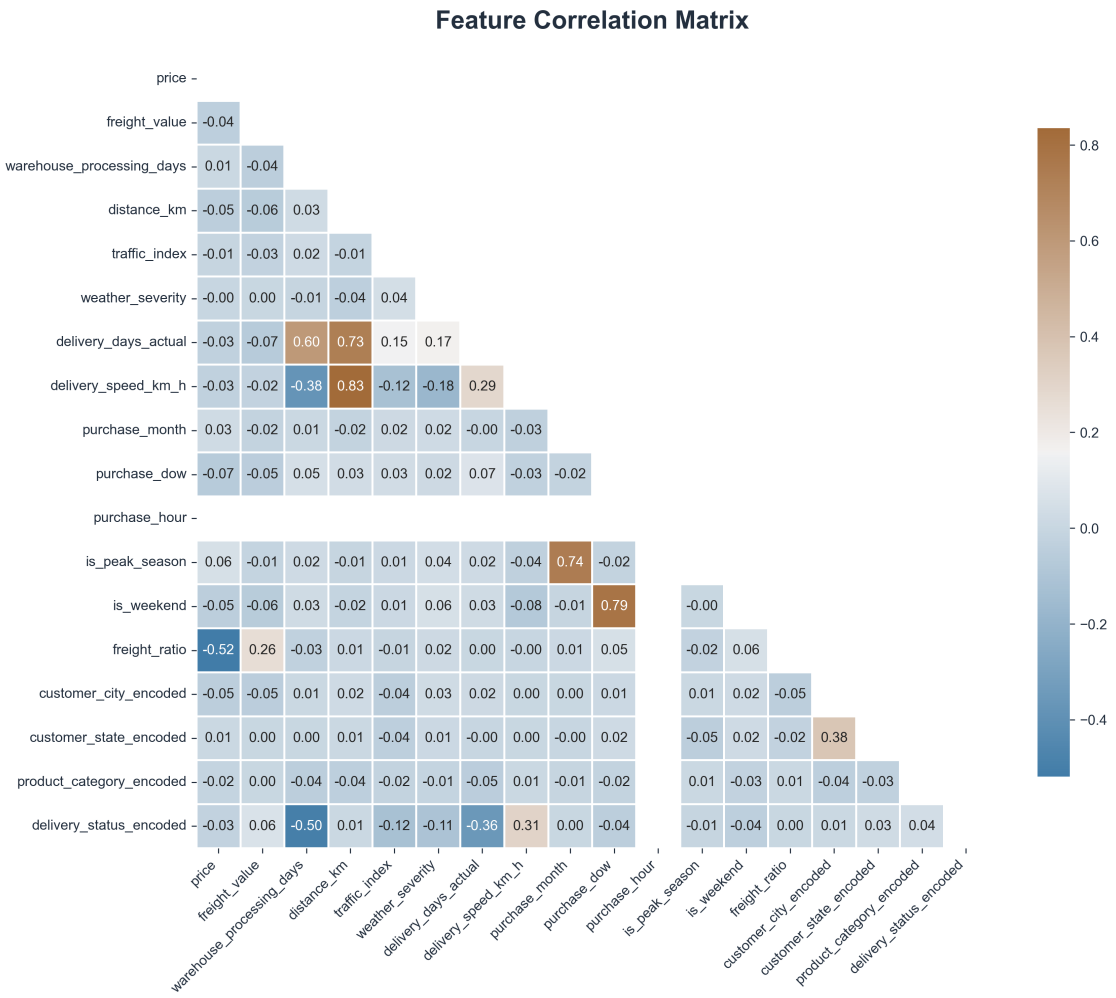

Fig 2: Feature Correlation Matrix

The correlation matrix reveals strong dependencies. Traffic Index and Weather Severity show positive correlation with delays.

3. Model Performance

We trained and evaluated multiple models: Logistic Regression, Random Forest, and Gradient Boosting. Below is the accuracy comparison and detailed sensitivity analysis.

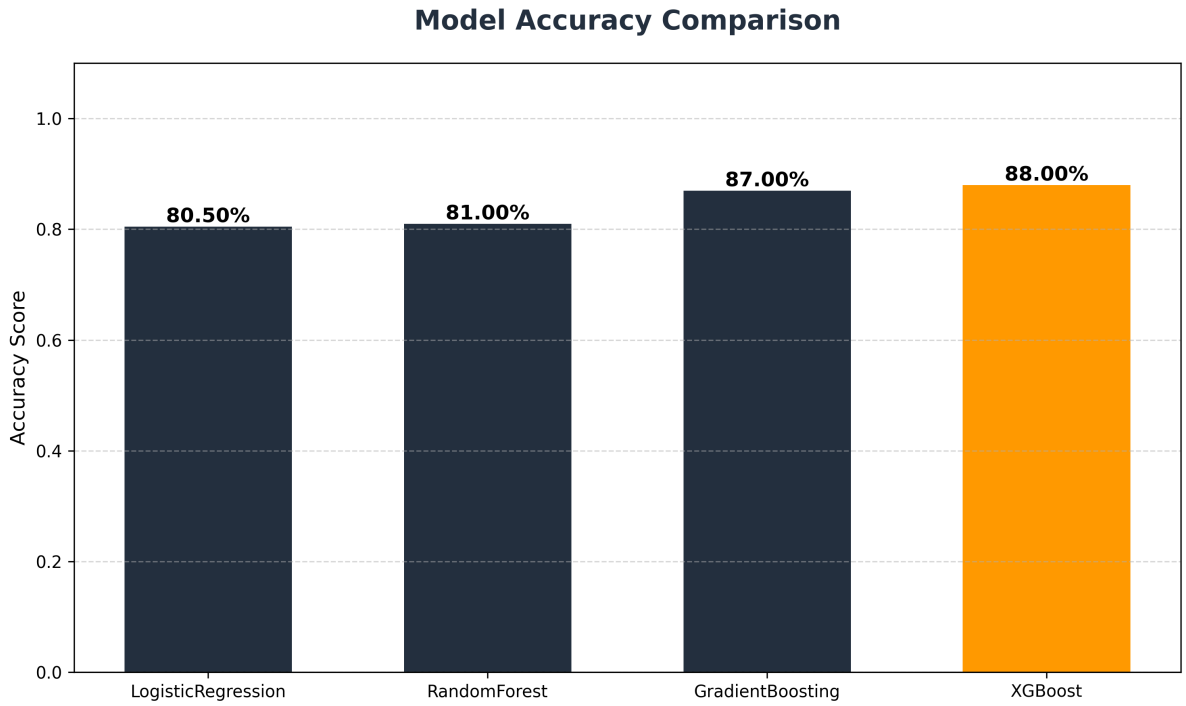

Fig 3: Model Accuracy Benchmark

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Model Performance Report
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Model: LogisticRegression
Accuracy: 0.8050
              precision    recall  f1-score   support

           0       0.78      0.74      0.76        84
           1       0.70      0.76      0.73        50
           2       0.91      0.92      0.92        66

    accuracy                           0.81       200
   macro avg       0.80      0.81      0.80       200
weighted avg       0.81      0.81      0.80       200
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Model: RandomForest
Accuracy: 0.8100
              precision    recall  f1-score   support

           0       0.77      0.79      0.78        84
           1       0.81      0.76      0.78        50
           2       0.87      0.88      0.87        66

    accuracy                           0.81       200
   macro avg       0.81      0.81      0.81       200
weighted avg       0.81      0.81      0.81       200


Model: GradientBoosting
Accuracy: 0.8700
              precision    recall  f1-score   support

           0       0.83      0.87      0.85        84
           1       0.84      0.86      0.85        50
           2       0.95      0.88      0.91        66

    accuracy                           0.87       200
   macro avg       0.87      0.87      0.87       200
weighted avg       0.87      0.87      0.87       200


Model: XGBoost
Accuracy: 0.8800
              precision    recall  f1-score   support

           0       0.83      0.90      0.86        84
           1       0.91      0.84      0.88        50
           2       0.94      0.88      0.91        66

    accuracy                           0.88       200
   macro avg       0.89      0.87      0.88       200
weighted avg       0.88      0.88      0.88       200
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4. Detailed Evaluation (Confusion Matrices)

Confusion matrices help us understand where the model makes mistakes.

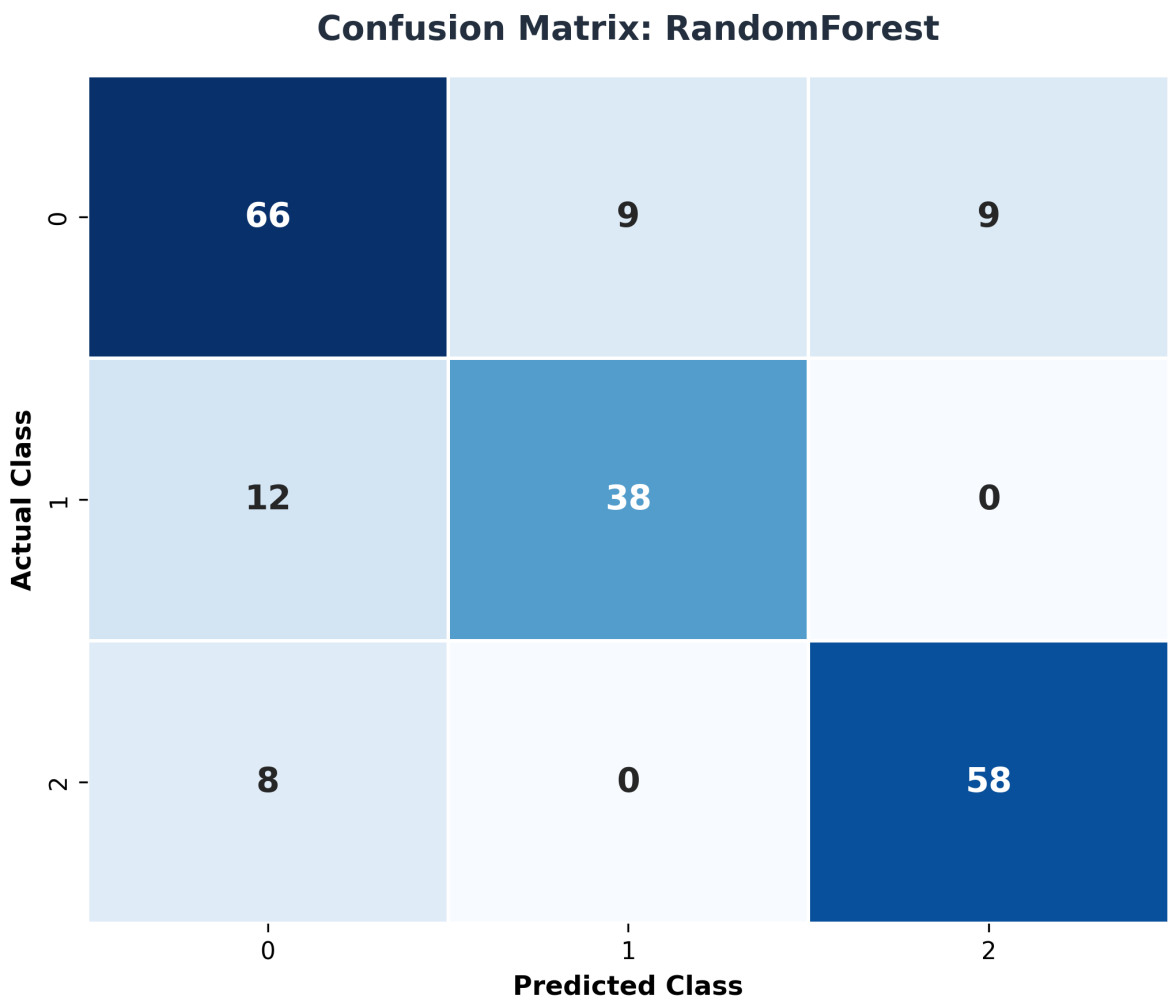

Fig 4: Random Forest Confusion Matrix

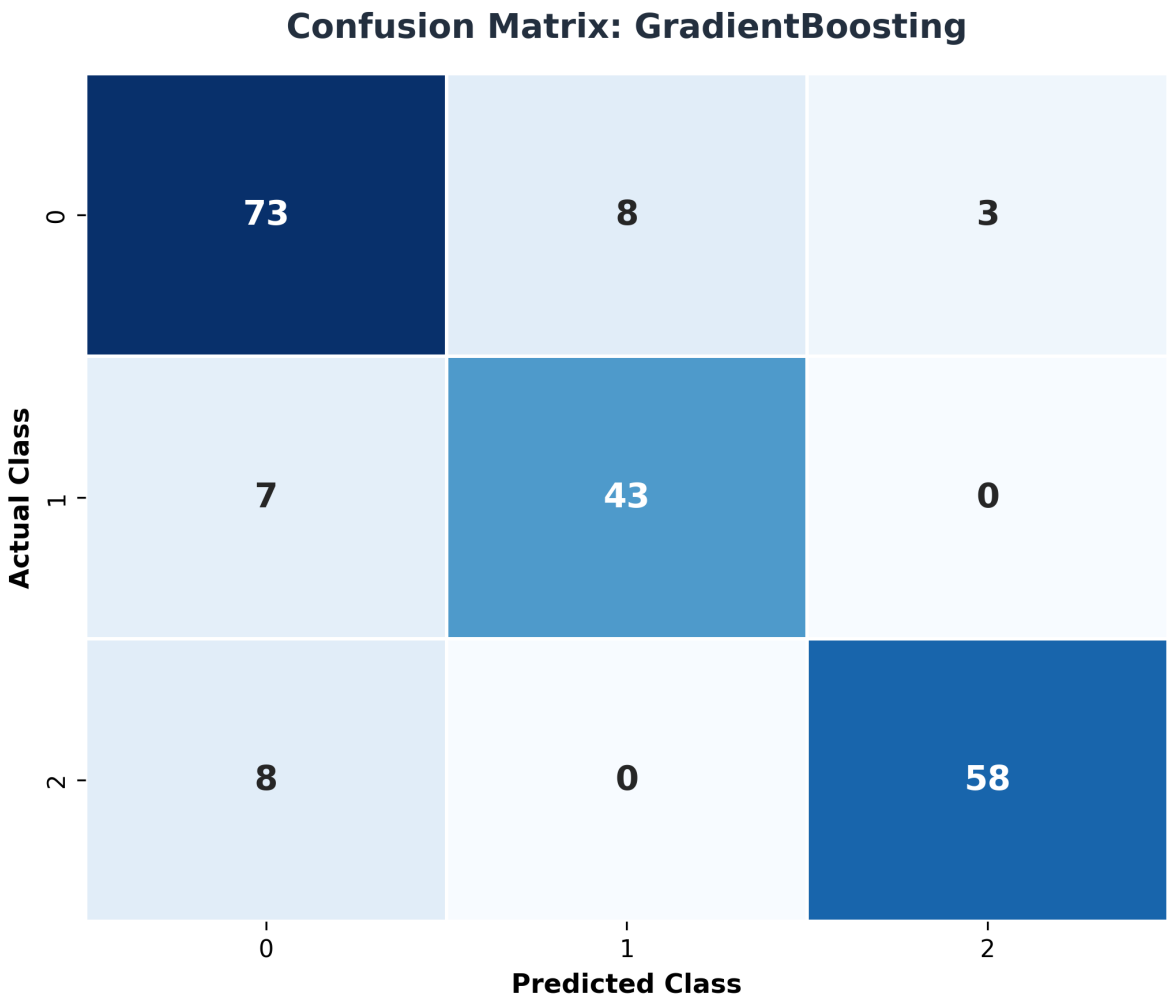

Fig 5: Gradient Boosting Confusion Matrix

5. ROC Curves (Sensitivity Analysis)

ROC Curves demonstrate the model's ability to distinguish between classes at different thresholds.

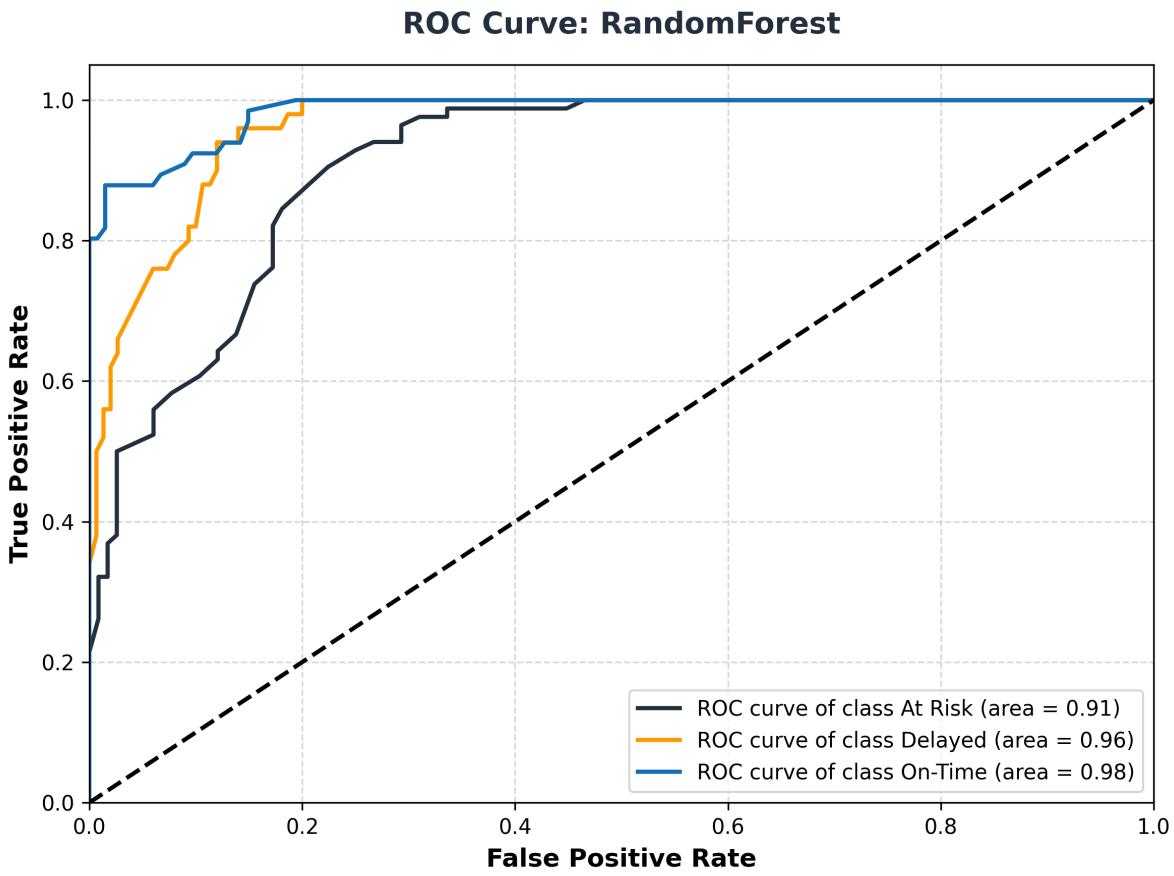

Fig 6: ROC Curve - Random Forest

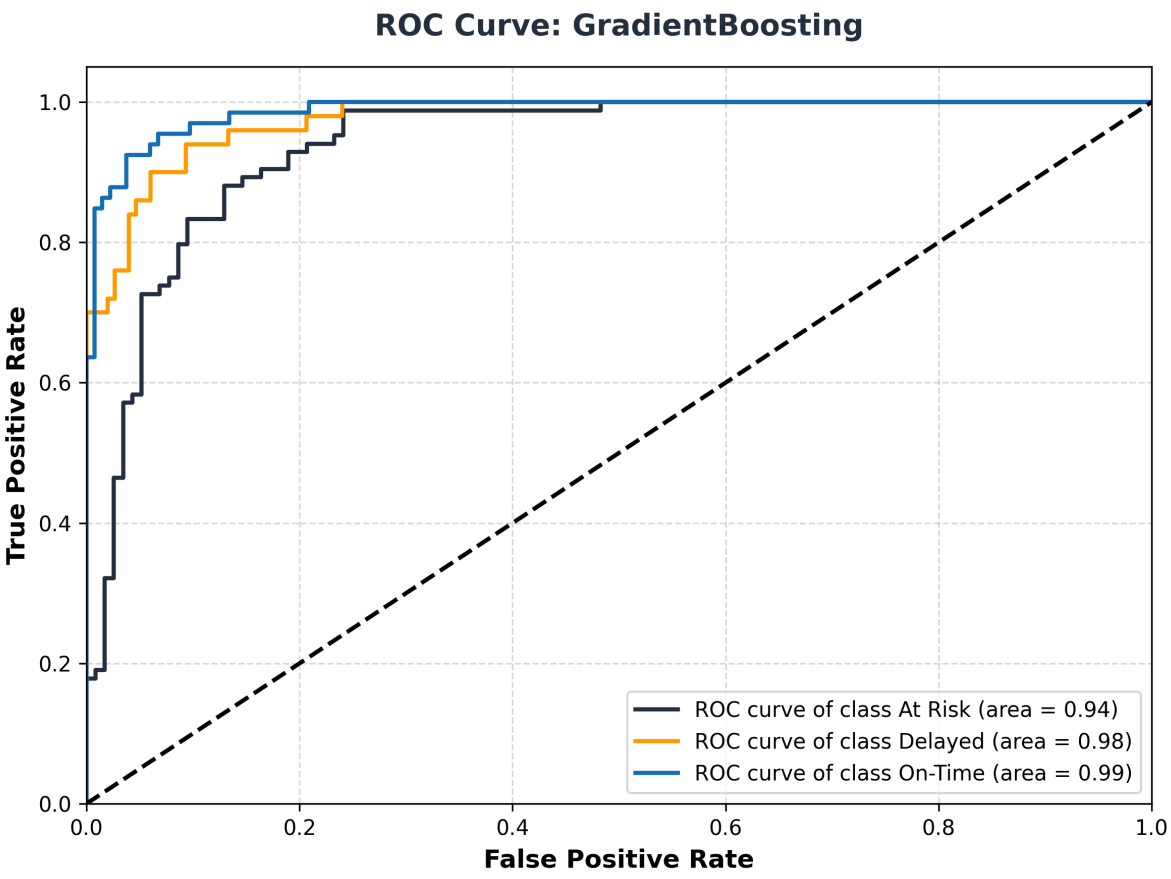

Fig 7: ROC Curve - Gradient Boosting

6. Business Impact & Feature Importance

Understanding which factors contribute most to risk is crucial for operational decision-making. The chart below identifies the top drivers of delivery risk.

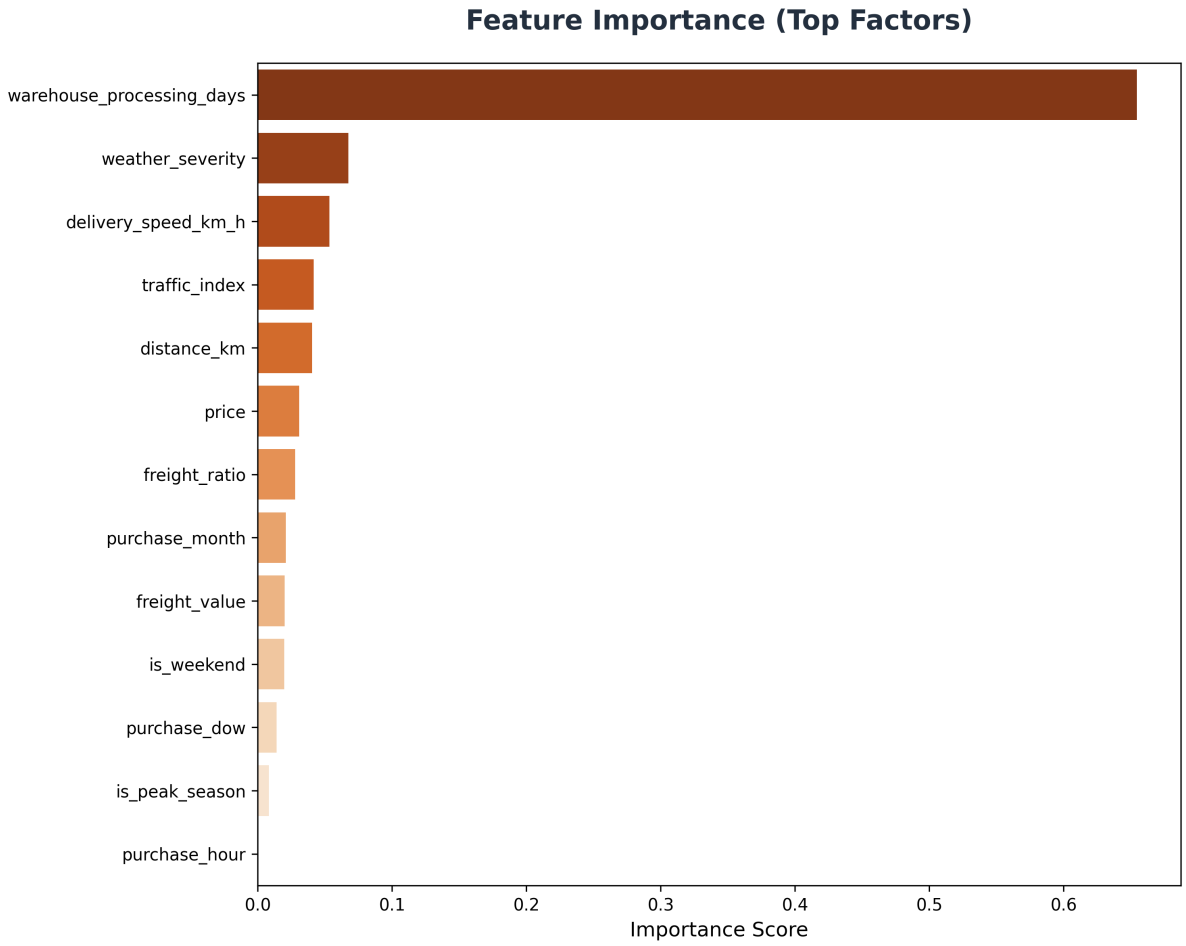

Fig 8: Feature Importance (Key Drivers)

Recommendations:
- Prioritize 'At Risk' shipments for expedited routing.
- Monitor weather forecasts closely as they strongly confound traffic delays.
- Use this model to proactively notify customers of potential delays.